

relative position between its two GNSS antennae to derive a heading solution that is an order of magnitude more accurate than a magnetic compass. It supports a variety of GNSS antennae that can be mounted on the host platform with a separation distance anywhere from a few centimetres to several metres.

"VN-360 is like no other product on the market, in that it provides a cost-effective solution for the difficult challenge of obtaining an accurate and reliable heading solution," stated John Brashear, president, VectorNav Technologies, in a recent press release.

He added, "VectorNav's GPS-Compass technology marks a turning point in the way we approach heading measurement and will improve the capabilities and performance of a variety of next-generation manned and unmanned systems."

Fancy an antenna, anyone.

Another product that was released recently is Acutime 360 from Trimble. It is a smartantenna with an integrated multi-GNSS receiver for high-accuracy and precise-timing applications. It includes support for GPS, GLONASS and BeiDou, and is Galileo-ready.

Acutime 360 has a tracking sensitivity of -160dBm and an acquisition sensitivity of -148dBm. It can operate in extreme temperatures (-40°C to +85°C) and hostile radio frequency environments typically encountered at wireless network transmitter sites, with a power input of less than one watt.

Highly minimised. Measuring merely 12.2mm x 16mm x 2.4mm, NEO-M8P is another very small, high-precision, GNSS real-time kinematic module, released by U-Blox, based on GPS and GLONASS satellite based navigation systems. Real-time kinematic algorithms are pre-integrated into the module. As a result, the size and weight are

significantly reduced and power consumption is five times lower than existing solutions.

In a recent press release, Daniel Ammann, executive director positioning and co-founder of U-Blox, said, "NEO-M8P lowers the barriers for innovative companies looking to develop equipment that need centimetre-level accuracy in many markets and applications such as unmanned aerial vehicles."

A wide-range augmented solution. BCM89774, a recent navigation solution launched by Broadcom, provides improved location and positioning while lowering power

support to Indian GAGAN augmentation system. According to a recent release, "Security mechanisms have also been embedded in FW 3.01. An anti-spoofing feature detects fake GNSS signals and a message integrity protection system prevents man-in-the-middle attacks. Another security function detects and suppresses jamming. A welcome feature of FW 3.01 is the ten per cent power reduction compared to earlier firmware versions."

Employment in wearables. All GNSS chipsets can easily be employed in handheld devices such as smartphones, tablets and palmtops.

But these require further miniaturisation when being used in wearables. Smartwatches have been the talk of town; selling like hot cakes, these essentially are mini computer systems.

With features like receiving and making phone calls, smartwatches have become a scaled-down version of your mobile phone. For these devices, we do not necessarily need accuracy of up to centimetres; a few metres is fine. Hence, GNSS chipsets supporting a single satellite system is fine when it comes to wearables.

"Asset tracking is an area where we have seen a lot of enquiries. Chips used in these wearables would usually be scaled-down versions of those used in vehicle tracking. In such cases, you could just use a GPS module without additional constellations as these really do not need that kind of accuracy," says Gulati.

With the boom in wearables, miniaturisation has been taken to the next level. Very soon we might have GNSS chipsets smaller than a nail with tracking accuracy in centimetres, giving a huge burst to GNSS-aided missile-guidance systems. The irony of such a helpful guidance system is that, it is growing by reducing the size. **EFY**

Major contributors to this report

- **Ashish Gulati**, country manager, Teit Wireless Solutions
- **M.N. Vidyashankar**, president, IESA
- **Rakesh C.R.**, field application engineer - GNSS Microelectronics and Solutions, Accord Software
- **Sudhir N.S.**, senior manager - GNSS technology, Accord Software

consumption in in-vehicle applications and reduces bill of materials cost for car makers. Offering simultaneous tri-band reception of all visible GNSS satellites including US GPS, European Galileo, Japanese QZSS, Russian GLONASS, Chinese BDS and global SBAS augmentation systems, Broadcom's BCM89774 sets itself apart as a wideband capture radio technology, delivering original equipment manufacturers one of the most accurate solutions available today.

GNSS firmware upgrade. With a firmware upgrade by U-Blox, for its earlier M8 module, U-Blox M8 FW 3.01 now also supports Galileo, in addition to GPS, GLONASS, BeiDou, QZSS and SBAS. It can track up to three constellations concurrently and makes use of all SBAS and QZSS augmentation systems at the same time.

In addition, with FW 3.01, U-Blox M8 now boasts of BeiDou acquisition sensitivity and adds